KENYA MEDICAL TRAINING COLLEGE
FACULTY OF CLINICAL SCIENCES
DEPARTMENT OF ORTHOPAEDIC AND TRAUMA MEDICINE

FINAL QUALIFYING EXAMINATION
FOR
CERTIFICATE IN ORTHOPAEDIC AND TRAUMA MEDICINE

PAPER: BASICS OF TRACTION TECHNIQUES

DATE: 6TH FEBRUARY, 2025 **TIME: 3 HOURS (9:00 – 12:00 NOON)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. ALL Multiple choice questions **MUST** be answered in **CAPITAL LETTERS**
4. Write your **EXAMINATION NUMBER** on all the answer sheets provided and on the question paper.
5. No other writing should be done on the question paper except your Examination Number.
6. Ensure that all examination answer scripts are handed in at the end of the examination
7. Ensure you sign the examination register provided

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SECTION 1: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. You have been asked to fix skeletal traction to a 20-year old patient with a minimally displaced fracture of proximal femur. Which of the following factors will you consider in fixing the weight?
   A. Orthopaedic bed
   B. Weight of the patient
   C. Surgeon's instruction
   D. Traction type
2. For the above patient in question 1, which is the appropriate site for inserting skeletal traction pin?
   A. Tibial tuberosity
   B. Calcaneal
   C. Malleolus
   D. Mid tibia
3. Patient J is 6-years old with bilateral fractures of mid shaft femur. Which of the following traction techniques will be suitable for conservative management?
   A. Spiral traction
   B. Gallows traction
   C. Hammock traction
   D. Dunlop traction
4. Patient X was applied skin traction for 3 days. During the ward round, the surgeon recommended for change of traction. Which of the following was a possible reason?
   A. Pressure sores
   B. Joint stiffness
   C. Infection
   D. Skin excoriation
5. Which one of the following is a least contraindication of skin traction?
   A. Sores
   B. Bruises
   C. Pain
   D. Wound
6. During your group discussion, you identified the purposes of traction. Which one of the following is a least indication of traction?
   A. Correction of deformities
   B. Relieving of muscle spasm
   C. Relieving of eodema
   D. Reducing of fractures
7. A 40-year old female patient sustained a displaced fracture of mid-shaft humerus. The patient was applied a hanging cast. Why was this cast appropriate?
   A. It is a shoulder immobilizer
   B. It permits swelling
   C. It is innocuous to the skin
   D. A form of traction by gravity

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8. Which of the following injuries would necessitate the use of Thomas splint?
   A. Scapular fractures
   B. Femoral fractures
   C. Shoulder dislocation
   D. Fractures forearm
9. Which of the following factors would determine the amount of force to be applied on traction technique?
   A. The site of injury
   B. Degree of displacement
   C. Age of the patient
   D. Type of traction
10. Why is lateral strap applied more posterior in skin traction?
   A. To encourage vertical pulling
   B. To avoid skin exfoliation
   C. To control rotation
   D. To relieve peronial nerve pressure
11. Which one of the following is the most ideal indication for head halter traction?
   A. Fractured lumber
   B. Neck fractures
   C. Clavicle fractures
   D. None of the above
12. During your visit to an orthopaedic ward, which of the following is least found on orthopaedic bed?
   A. Firm mattress
   B. Fracture board
   C. Traction Cords
   D. Balcum beam
13. Which of the following pins is least used during skeletal traction on patients?
   A. Steinman's pin
   B. Denham pin
   C. Shanz screw
   D. Introducer pin
14. Which one of the following would you least put a skeletal traction pin on management of patients with fractures?
   A. Olecranon process
   B. Skull
   C. Metatarsals
   D. Distal femur

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15. Which of the following injuries would require application of a lateral upper femoral traction?
    A. Fracture grater trochanter
    B. Separation of sacro-iliac joint
    C. Acetabulum fracture
    D. Central hip dislocation
16. Patient V had a displaced fracture of mid-shaft tibia. Which of the following pin sites will you fix a skeletal traction for conservative management?
    A. Mid 1/3 tibia
    B. Distal femur
    C. Calcaneum
    D. Tibial tuberosity
17. Which of the following is would least be found in sliding traction?
    A. Bucks extension skin traction
    B. Perkins traction
    C. Pelvic traction
    D. Fixed traction
18. Patient N had a displaced fracture of mid 1/3rd femur. The patient was fixed a Bohler Braun frame, which of the following is the advantage of this?
    A. Allows easy carrying of patients
    B. Allows easy reduction of fractures
    C. Allows oedema to subside easily
    D. Allows cleaning of patients easily
19. Which of the following statements best explains how manual traction works?
    A. Use of pulleys and weights
    B. One force pulling in the opposite direction
    C. Two points forces pulling in the same directions
    D. Balanced three point forces
20. While applying skeletal traction on the tibial bone, what should be the angle position between the drilling machine and the bone?
    A. 30°
    B. 180°
    C. 45°
    D. 90°
21. What is the role of an orthopaedic trauma technician/technologist after the application of skeletal traction to a patient?
    A. Train the patient on how to use crutches
    B. Discharge the patient home
    C. Give social economic support to the patient
    D. Monitor on patient progress on traction

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22. During ward round, you found a patient on head halter traction. Which of the following is an indication?
    A. Treatment of spinal injuries
    B. Management of lower limb injuries
    C. Upper limb deformities
    D. Management of cervical injuries
23. Which of the following tractions is prone to pin tract infection?
    A. Gallow's traction
    B. Bryant's traction
    C. Sliding traction
    D. Skeletal traction
24. When you are fixing the distal femur skeletal traction, what would be the direction of fixing the pin?
    A. Postero-anterior
    B. Lateral-medial
    C. Antero-posterior
    D. Medial-lateral
25. Which one of the following complications would least be caused by skeletal traction?
    A. Loose Steinmann pic
    B. Pin tract infection
    C. Osteomyelitis
    D. Malignant change
26. How will you identify a sequestrum bone on a plain radiograh of a patient with chronic osteomyelitis secondary to skeletal traction?
    A. Sinus filled bone
    B. Cloud like bone
    C. Light filled bone
    D. Dark like bone
27. Which of the following complications would least be caused by Bryant's traction?
    A. Nerve injury
    B. Arterial injury
    C. Hip dislocation
    D. Bone infection
28. In hip dislocation of an adult patient, for how long would the traction be maintained?
    A. 4-6 weeks
    B. 8-12 weeks
    C. 6-8 weeks
    D. 3 weeks
29. Which of the following injuries would be necessary to apply Dunlop traction?
    A. Epiphyseal injuries of the upper extremity
    B. Supracondylar fractures of the humerus
    C. Elbow stiffness and myositis ossificans
    D. Supracondylar fractures of femur in children

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30. Patient M fell and sustained prolapsed intervertebral disc. Which of the following tractions would be appropriate for him?
    A. Bucks traction
    B. Hamilton Russel traction
    C. Pelvic traction
    D. Perkin's traction
31. Which of the following parts of the lower limb would you palpate for common peroneal nerve?
    A. Neck of femur
    B. Head of fibula
    C. Head of tibia
    D. Distal femur
32. Which of the following is the use of extension bars on orthopaedic bed?
    A. To rest the weight
    B. To rotate the weight
    C. To double the weight
    D. To suspend weight
33. Which of the following care will reduce the risk of pin tract infection?
    A. Frequent change of traction
    B. Covering the pins with a clean cloth
    C. Sterility of the traction procedure
    D. Elevation of the limb on traction
34. A patient presented with a displaced fracture of mid-shaft femur. Which of the following medication would be administered to manage pain?
    A. Ampiclox
    B. Ethanol
    C. Tramadol
    D. Prednisolone
35. Which of the following physical assessment is done to a patient before insertion of traction pins?
    A. Cardiovascular assessment
    B. Skeletal assessment
    C. Neurovascular assessment
    D. Respiratory assessment
36. In which of the following conditions will you use a hip spica table?
    A. Cervical injuries in children
    B. Thoracic injuries in adults
    C. Upper limb injuries in children
    D. Femoral injuries in children

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37. Patient R had spondylolisthesis of lumber 4. Which of the following tractions would you apply?
    A. Pelvic binder
    B. Spinal skeletal traction
    C. Milwauke traction
    D. Spinal traction
38. Which one of the following is the importance of pulleys in traction?
    A. Provide non mechanical advantage by force decrease
    B. A complex machine used to increase force
    C. Single pulleying used not to change the pull direction
    D. Provide mechanical advantage by increasing force efficiency
39. Which of the following condition would require bilateral skin traction as a mode of management?
    A. Displaced left pubis ramus
    B. Correction Lordosis
    C. Prolapsed intervertebral disc
    D. Thoracic injury in adults
40. Which one of the following conditions would least require skull traction?
    A. Fracture cervical bone
    B. C_4 subluxation
    C. Dislocation of the cervical bone
    D. Lumbar 2 subluxation

SECTION 2: SHORT ANSWER QUESTIONS (40 MARKS)

41. Enumerate (5) five complications of skin traction **(5 marks)**
42. Outline (5) five principles of tractions **(5 marks)**
43. State (5) five indications of internal fixation **(5 Marks)**
44. Highlight (5) five essential components of skeletal traction **(5 marks)**
45. Enumerate (5) five sites for insertion of skeletal traction **(5 marks)**
46. List (5) five general complications of patients on traction **(5 marks)**
47. Differentiate between skin and skeletal traction **(5 marks)**
48. List (5) five components of an ideal orthopaedic bed **(5 marks)**

SECTION 3: LONG ANSWER QUESTIONS (20 MARKS)

49. A fourteen years old girl fell while evading a stray dog, she slid and fell down. She presented with pain, swelling and inability to use the left lower limb.
    a. What is the possible diagnosis? **(2 marks)**
    b. Discuss the possible traction that would be applied on her under the following headings.
        i. Type of traction **(2 marks)**
        ii. Requirements and preparations **(4 Marks)**
        iii. Procedure of application **(10 Marks)**
        iv. Post traction advice **(2 marks)**

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